Sentinel Early Warning Detection System

Complete Architecture & Design Document

Version: 2.0
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Project: Azentio AI Hackathon - Problem Statement 4
Team Branch: feat/pranav

1. Executive Summary

Sentinel is an AI-powered Early Warning System (EWS) that continuously evaluates customer repayment behaviour, financial exposure, and emerging risk signals to identify potential credit deterioration before serious delinquency occurs. Built for the Azentio AI Hackathon (Problem 4), it covers 8 early warning scenarios, 21 risk signals, and delivers a complete analyst workflow from data ingestion through watchlist triage to regulatory reporting.

The system achieves:

- 97.5% accuracy within one tier of ground truth
- 75.8% catch rate for High/Critical risk customers
- 0.0% false alarm rate on low-risk customers
- AUC 0.96 for 1-month delinquency forecast
- AUC 0.89 for 3-month delinquency forecast

2. Problem Statement

2.1 Business Context

Banks face significant losses when credit deterioration goes undetected until customers are already in serious delinquency. Traditional approaches relying on periodic reviews and hard DPD thresholds miss the early, subtle signals of financial stress - declining income, increasing leverage, partial payments, and cross-lender distress.

2.2 Challenge Requirements

Design and build an Early Warning Detection System that satisfies 5 Level 1 capabilities:

#		
L1-1	**Data Ingestion**	Ingest and validate multi-source ba
L1-2	**Signal Detection**	Detect 8 early warning scenarios co
L1-3	**Warning Score**	Compute a composite risk score (0-1
L1-4	**Watchlist**	Generate and manage a prioritised w
L1-5	**Analyst Review**	Provide a full analyst workflow wit

2.3 Early Warning Scenarios

Scenario	Code	Description
S1	Repayment Deterioration	Increasing missed/late payments, wo
S2	Declining Income	Sustained drop in inflows, salary l
S3	Increasing Exposure	Rising utilisation, new credit, deb
S4	Temporary Exception	Isolated missed payment with subseq
S5	Roll-Forward Risk	Progressive migration across DPD bu
S6	SME Stress	Seasonal cash-flow stress, sector-s
S7	Missing History	Thin files, incomplete data, bureau
S8	Portfolio Clusters	Emerging risk concentration by geog

2.4 Target Setting

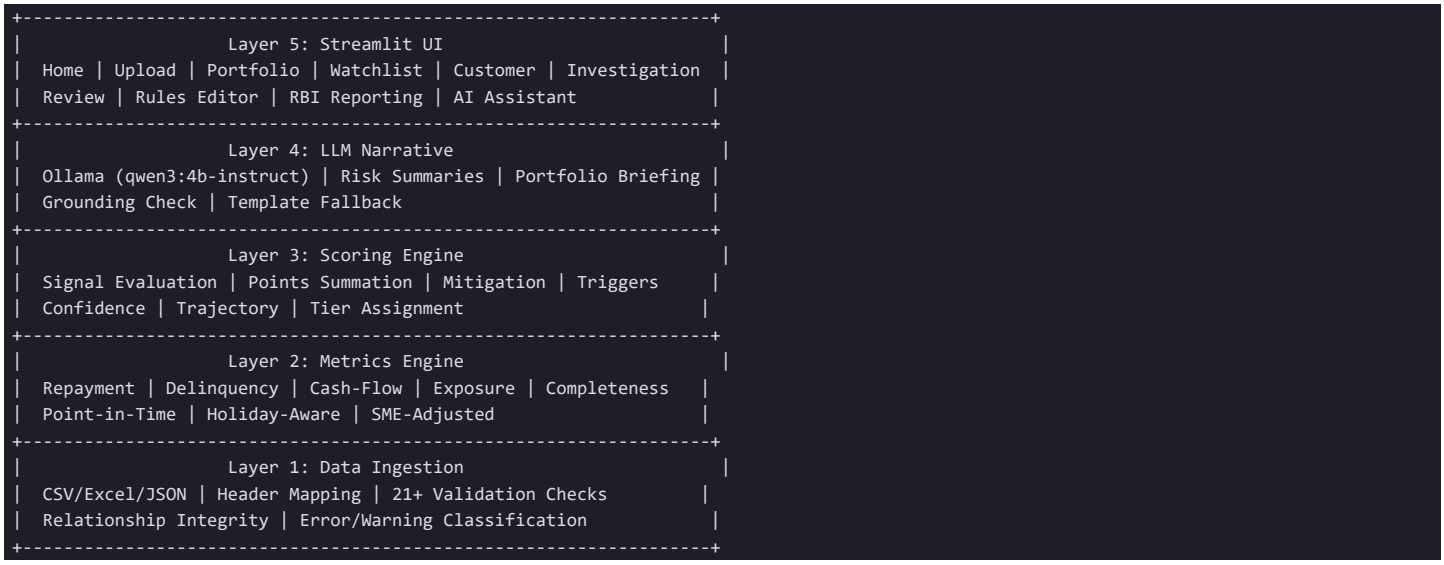
- Geography: UAE (AED currency, Emirates-based distribution)
- As-of Date: 2026-09-30
- History Window: 24 months (October 2024 - September 2026)
- Customer Segments: Individual (retail) and SME
- Portfolios: Two synthetic bank datasets for validation

3. Solution Overview

3.1 Design Principles

1. Rules-first, ML-second: Transparent rule-based scoring is the primary system. ML provides a "second opinion" and forecasting layer, never overriding rules.
2. Absolute anchoring: Signal thresholds use absolute values (e.g., DPD >= 30), not percentile-based, ensuring consistency across portfolio sizes.
3. Two-stage compute: Expensive metrics computation is cached; cheap scoring re-runs instantly on rule changes.
4. LLM as narrator, not calculator: The LLM generates natural-language summaries but never computes scores or changes decisions.
5. Audit everything: Frozen alerts, append-only decisions, intervention history - nothing is ever deleted or mutated.

3.2 Five-Layer Architecture



4. System Architecture

4.1 Module Map

The sentinel/ package contains 14 Python modules, each with a single responsibility:

Module		
`ingest.py`	Data loading, validation, cleaning	`load_and_validate()`
`metrics.py`	Point-in-time metric computation	`compute()`, `_repayment_metrics()`
`scoring.py`	Signal evaluation and tier assignme	`score()`
`forecast.py`	LightGBM delinquency prediction	`train_delinquency_model()`, `predi
`ml.py`	ML second opinion with SHAP	`build_ml_second_opinion()`, `expla
`llm.py`	Local LLM integration	`summarise_customer()`, `portfolio_
`insights.py`	Roll-rates, clusters, recovery prof	`roll_rate_matrix()`, `emerging_clu
`store.py`	DuckDB persistence and audit	`publish_alerts()`, `record_decisio
`correlation.py`	Multi-signal pattern detection	`analyse()`
`interventions.py`	Intervention recommendations	`recommend()`
`priority.py`	Composite priority scoring	`score()`, `quadrants()`
`rbi.py`	RBI SMA regulatory reporting	`classify_sma()`, `compliance_summa
`trends.py`	Historical trend analysis	`compute_history()`, `extended_traj

4.2 Data Flow



4.3 Two-Stage Compute Model

Stage 1 - Metrics (Heavy, Cached):

- Runs once per dataset + as-of-date combination
- Computes all 21 metric values per customer using vectorised pandas
- Handles holiday-aware grace periods, SME seasonal adjustments

- Result cached in st.session_state (~0.4s for 600 customers)

Stage 2 - Scoring (Cheap, Real-time):

- Re-runs on every rule change, What-If scenario, or parameter edit
- Maps metrics to signal strengths using absolute thresholds
- Computes weighted sum, mitigation, triggers, confidence, trajectory
- Result: scored DataFrame with tier, score, reasons, evidence (~0.5s)

5. Data Architecture

5.1 Input Tables

Table	Key Columns	Grain
customers	customer_id, customer_type, custome	One row per customer
facilities	facility_id, customer_id, product_c	One row per facility
repayments	repayment_id, facility_id, customer	One row per instalment
financials	month, customer_id, salary_credit,	Monthly per customer
exposure	snapshot_date, customer_id, revolvi	Monthly snapshot per customer
delinquency	snapshot_date, facility_id, custome	Monthly snapshot per facility
bank_holidays	holiday_date, holiday_name, holiday	Reference table
exception_evidence	evidence_id, customer_id, event_dat	Event-level evidence

5.2 Synthetic Portfolios

Portfolio 1: Synthetic Gulf Bank

- 600 customers (420 individual, 180 SME)
- Retail-led portfolio, 0.2% NPL rate
- Default profile, AED currency

Portfolio 2: Northern Emirates Commercial Bank

- 1,200 customers (660 individual, 540 SME)
- SME-heavy mid-market bank, 5.6% NPL rate
- Heavier stress archetypes including LEGACY_NPL
- 3 planted risk clusters: CLU-SME-LOGI-NORTH, CLU-RET-TRADE-NORTH, CLU-SME-REST-AJM

5.3 Data Quality Testing

The system includes a dirty data variant with 73 deliberately injected issues across 10 defect types:

Defect T		
MISSING_MANDATORY	~8	customers, facilities
INCONSISTENT_FORMAT	~7	customers, repayments
INVALID_DATE	~8	facilities, repayments
DUPLICATE_RECORD	~7	customers, facilities
ORPHAN_REFERENCE	~7	facilities, repayments
NEGATIVE_AMOUNT	~8	facilities, repayments
CURRENCY_MISMATCH	~7	facilities
INCONSISTENT_STATUS	~7	facilities
RELATIONSHIP_MISMATCH	~7	repayments
OUTLIER	~7	facilities

6. Data Ingestion & Validation Pipeline

6.1 File Loading (ingest.py)

The ingestion module supports three file formats:

- CSV: Direct pandas read with UTF-8 encoding
- Excel (.xlsx): Reads every sheet, detects table type from sheet name
- JSON: Handles both array-of-objects and keyed-object structures

Table Detection: Uses filename and sheet name hints:

- customer / cust -> customers
- facilit / loan -> facilities
- repay / payment -> repayments
- financial / cashflow -> financials
- exposure / bureau -> exposure
- delinq / dpd / arrear -> delinquency
- holiday -> bank_holidays
- evidence / exception -> exception_evidence

6.2 Header Normalisation

All headers are lowercased and snake_cased, then mapped through an alias dictionary:

```
loan_id, account_id          -> facility_id
product_type                 -> product_name
outstanding, balance         -> outstanding_balance
instalment, emi              -> instalment_amount
payment_amt                  -> amount_paid
```

6.3 Validation Checks (21+)

#	Check	Severity
1	Missing mandatory columns	ERROR
2	Null values in required fields	ERROR
3	Invalid date formats	ERROR
4	Future dates beyond as-of	WARNING
5	Negative monetary amounts	ERROR
6	Duplicate primary keys	ERROR
7	Orphan foreign key references	ERROR
8	Invalid enum values (status, type)	WARNING
9	Text in numeric columns	ERROR
10	Currency mismatches	WARNING
11	Maturity before disbursement	WARNING
12	Outstanding exceeds sanctioned	WARNING
13	Payment exceeds amount due	WARNING
14	Zero or negative interest rates	WARNING
15	Facility-customer relationship inte	ERROR
16	Repayment-facility relationship int	ERROR
17	Status consistency (closed + outsta	WARNING
18	DPD bucket vs days_past_due consist	WARNING
19	Outlier detection (statistical)	WARNING
20	Missing financial coverage gaps	WARNING
21	Bureau score range validation	WARNING

Output: (clean_tables, issues, summary) where issues carry ERROR (row rejected) or WARNING (normalised and kept) severity.

7. Metrics Engine

7.1 Overview (metrics.py)

The metrics engine computes point-in-time values for every customer at a given as-of date. All metrics are computed using vectorised pandas operations for performance.

7.2 Repayment Metrics

Metric	Description	Logic
`late_or_missed_instalments`	Count of months with late/missed pa	Per-month (not per-instalment) in t
`partial_payments`	Count of partial payments	amount_paid > 0 but < amount_due
`isolated_delay_recovered`	Boolean: single late month followed	Exactly 1 late month in window AND
`evidence_verified`	Boolean: exception evidence exists	Matches customer_id + verified_flag
`consecutive_missed_months`	Consecutive months of missed paymen	Used for hard trigger (>=2 -> at le

7.3 Delinquency Metrics

Metric	Description
`current_dpd`	Current days past due (max across f
`dpd_increase_month_ends`	Count of consecutive monthly DPD in
`months_since_cure`	Months since last transition from 3
`max_dpd_12m`	Maximum DPD in trailing 12 months
`dpd_history`	Full DPD time series (used by trend

7.4 Cash-Flow Metrics

Metric	Description	SME Adjustment
`inflow_change_3m_vs_prior_3m_pct`	3-month average inflow vs prior 3 m	SMEs use YoY comparison to account
`consecutive_inflow_declines`	Count of consecutive monthly inflow	SMEs use YoY ratio instead of MoM
`months_without_salary`	Months with no salary credit (salar	N/A for SMEs
`avg_balance_change_pct`	Average balance change over period	Same for both
`returned_payments`	Count of returned/bounced payments	Same for both

7.5 Exposure Metrics

Metric	Description
`revolving_utilisation_pct`	Current revolving balance / limit
`utilisation_change_pp`	Percentage point change in utilisat
`debt_burden_ratio_pct`	Total obligations / income (individ
`new_facilities_or_enquiries`	New facilities + bureau enquiries i
`exposure_growth_minus_income_growt`	Exposure growth rate minus income g
`bureau_external_max_dpd`	Max DPD at other lenders
`bureau_score_change`	Bureau score change over period
`bureau_no_hit`	Boolean: no bureau record found

7.6 Data Completeness Score

```
DCS = 0.35 x history_score + 0.30 x coverage_score + 0.15 x bureau_score + 0.20 x income_visible
```

Where:

- history_score = min(months_of_history / 12, 1.0)
- coverage_score = 1.0 - (missing_financial_months / expected_months)
- bureau_score = 0 if bureau_no_hit, else 1.0
- income_visible = 1.0 if salary_transfer or has financials, else 0.5

8. Risk Scoring Engine

8.1 Signal Definitions (scoring.py)

21 signals organised into 5 categories:
Repayment Signals (RBDI):

Signal				
RS01	Late or missed instalments	30	>= 2	>= 4
RS02	Worsening DPD trend	15	>= 2 consecutive rises	>= 4
RS03	Current arrears	15	DPD >= 30	DPD >= 90 (bucket-scaled)
RS04	Partial payments	10	>= 2	>= 4
RS05	Isolated delay recovery (mitigant)	-10	isolated_delay_recovered=True	x1.5 if verified
RS06	Recent cure	8	months_since_cure <= 6	months_since_cure <= 3

Cash-Flow Signals (CFSI):

Signal				
RS07	Sustained inflow decline	20	decline >= 15%	>= 30%
RS08	Consecutive inflow declines	8	>= 3 months	>= 5
RS09	Salary/income stopped	15	>= 2 months no salary	>= 4
RS10	Balance erosion	10	decline >= 30%	>= 60%
RS11	Returned payments	10	>= 2	>= 4

Exposure Signals (LPI):

Signal				
RS12	High revolving utilisation	12	>= 75%	>= 90%
RS13	Utilisation spike	10	increase >= 20pp	>= 40pp
RS14	Debt burden ratio	15	>= 50% (individuals only)	>= 70%
RS15	New credit accumulation	12	>= 3 new facilities/enquiries	>= 5
RS16	Exposure outpacing income	10	gap >= 15pp	>= 30pp

Bureau Signals:

Signal				
RS17	External delinquency	12	bureau_max_dpd >= 30	>= 60
RS18	Bureau score drop	6	drop >= 50 points	>= 100

Data Quality Signals (weight = 0, affects confidence only):

Signal		
DQ01	Insufficient history	Reduces confidence
DQ02	Missing financial months	Reduces confidence
DQ03	Bureau no-hit	Reduces confidence

8.2 Scoring Pipeline

```
For each customer:

1. SIGNAL EVALUATION
   - For each enabled signal, evaluate metric against threshold
   - strength = 0.0 (not firing), 0.5 (threshold met), 1.0 (severe met)

2. RAW SCORE COMPUTATION
   - raw_score = SUM(weight x strength) for positive-weight signals
   - Capped at 100

3. MODULE INDICES
   - RBDI = (RS01-RS06 earned points) / (RS01-RS06 available) x 100
   - CFSI = (RS07-RS11 earned points) / (RS07-RS11 available) x 100
   - LPI  = (RS12-RS16 earned points) / (RS12-RS16 available) x 100
   - BUREAU = (RS17-RS18 earned points) / (RS17-RS18 available) x 100

4. MITIGATION (RS05 only)
   - If isolated_delay_recovered AND (CFSI < 50 AND LPI < 50):
     - Apply RS05 weight (-10) x strength
     - If evidence_verified: multiply by 1.5 (= -15)
   - If CFSI >= 50 OR LPI >= 50: mitigation blocked

5. RISK SCORE = clip(raw_score + mitigation, 0, 100)

6. TIER ASSIGNMENT
```

```
   - Green:  0-24
   - Watch: 25-44
   - Amber: 45-69
   - Red:   70-100

7. HARD TRIGGERS (override tier upward only)
   - DPD >= 90           -> minimum Red
   - DPD >= 30           -> minimum Amber
   - consecutive_missed >= 2 -> minimum Amber

8. CONFIDENCE
   - High:   DCS >= 75
   - Medium: DCS >= 40
   - Low:    DCS < 40
   - LOW CONFIDENCE OVERRIDE: Low + no hard evidence -> capped at Watch (OV03)

9. TRAJECTORY (vs score 3 months ago)
   - New:           no prior score
   - Deteriorating: score increased >= 10 (adds +10 to score)
   - Improving:     score decreased >= 10 (subtracts -10 from score)
   - Recovering:    previously Amber/Red, now lower
   - Stable:        change < 10

10. OUTPUT: tier, score, confidence, trajectory, reasons[], evidence{}, module_indices{}
```

8.3 Score Bands

Band	Score Range	Status	Recommended Action
Green	0-29	NORMAL	Routine monitoring
Watch	30-59	WATCHLIST	RM review; soft-touch contact
Amber	60-79	ACTION_REQUIRED	Contact customer; consider restruct
Red	80-100	ESCALATE	Credit committee / collections refe

9. Machine Learning Pipeline

9.1 Delinquency Forecast (forecast.py)

Objective: Predict whether a customer currently below 30 DPD will breach 30+ DPD within 1 month and 3 months.
Training Approach:

- Point-in-time panel construction: metrics recomputed at 17 historical month-ends
- Creates ~8,800 training rows (600 customers x ~15 usable snapshots)
- Labels derived from actual future delinquency outcomes

Model Configuration:

LightGB

```
- n_estimators: 300
- max_depth: 4
- learning_rate: 0.03
- min_child_samples: 40
- subsample: 0.8
- colsample_bytree: 0.8
- scale_pos_weight: auto (adjusted for class imbalance)
```

Validation:

- Out-of-time split: last 3 monthly snapshots held out
- No data leakage: all features computed from data available at prediction time

Results:

- AUC 0.96 (1-month horizon)
- AUC 0.89 (3-month horizon)
- Top-10% probability capture: 90% (1m), 67% (3m)

Output per customer:

- p_1m: Probability of 30+ DPD within 1 month
- p_3m: Probability of 30+ DPD within 3 months
- Top-3 SHAP feature drivers

Red-Flag Rule: If P(30+ DPD 3m) >= 20% AND customer is Green/Watch -> flagged as "silent stress"

9.2 ML Second Opinion (ml.py)

Objective: Predict whether a customer will be Amber or Red (per the rules engine) in 3 months.
Enhancements over baseline:

Enhanc

SMOTE Oversampling	Balances minority class (Amber/Red)
Scale Pos Weight	Additional class imbalance handling
Stratified 5-Fold CV	Cross-validation preserving class r
Threshold Tuning	Optimises classification threshold
Interaction Features	6 pairwise interactions + aggregate

Interaction Features:

- late_x_dpd: late_or_missed x current_dpd
- inflow_x_balance: inflow_change x balance_change
- util_x_burden: utilisation x debt_burden
- missed_x_inflow: consecutive_missed x inflow_decline
- dpd_trend_x_exposure: dpd_increase x exposure_growth
- salary_x_balance: salary_stopped x balance_erosion
- repay_sum, cashflow_sum, exposure_sum: module aggregates
- n_signals_firing: count of metrics above non-zero thresholds

Output per customer:

- P(deterioration 90d): probability of reaching Amber/Red
- SHAP waterfall chart showing top feature contributions

9.3 Roll-Rate Analysis (insights.py)

Markov Transition Matrix:

- Built from ~19,000 facility-month delinquency snapshots
- States: Current -> 1-30 -> 31-60 -> 61-90 -> 91-180 -> 180+
- Computed per portfolio x bucket
- Each customer gets the highest p_roll across their facilities

Emerging Cluster Detection:

- Wilson confidence interval test on segment-level Amber/Red rates
- Identifies segments with statistically elevated risk vs portfolio average
- Reports lift ratio and cluster size

Recovery Profiles:

- Tracks customers who transitioned from Amber/Red to Green/Watch
- Analyses recovery time distribution and common recovery patterns

10. LLM Integration

10.1 Architecture (llm.py)

Runtime: Ollama (local inference, no cloud dependency)
Model Hierarchy:

1. Primary: qwen3:4b-instruct
2. Fallback: llama3.2:3b
3. Final fallback: Template-based summaries (always available)

Configuration:

```
tempera
max_tokens: model-dependent
think: false (disable chain-of-thought for speed)
```

10.2 Use Cases

Feature	Description	Grounding
Customer Summary	Natural-language risk narrative for	All numbers > 12 must appear in evi
Portfolio Briefing	Morning briefing summarising portfo	Grounded against actual KPIs
AI Assistant	Multi-turn chat for analyst questio	Conversation grounded in customer e

10.3 Guardrails

1. Number Grounding Check: Any number > 12 in LLM output must exist in the evidence provided. Ungrounded numbers trigger a fallback to template summary.
2. LLM Never Computes Scores: The scoring engine is the sole source of truth. LLM output is purely narrative.
3. LLM Never Changes Decisions: Analyst decisions are recorded through the structured workflow, not through chat.
4. Template Fallback: If Ollama is unavailable or grounding fails, a template-based summary is generated from the scored data.

11. Advanced Analytics Modules

11.1 Risk Correlation (correlation.py)

Detects multi-signal co-occurrence patterns that indicate systemic stress:

Pattern			
CORR01	Cash-flow driven delinquency	CFSI > 30 AND RBDI > 30	Income stress causing payment defau
CORR02	Leveraging up under stress	LPI > 30 AND (CFSI > 30 OR RBDI > 3	Borrowing to cover shortfalls
CORR03	Cross-lender distress	BUREAU > 30 AND (RBDI > 30 OR LPI >	System-wide financial stress
CORR04	Full deterioration	3+ modules > 30	Broad-based credit deterioration

Combined Risk Factor:

```
base =
combined_risk_factor = base + 0.15 x count(matched patterns)
```

Overlap Warning: Flags signal pairs that may double-count the same underlying risk (e.g., RS01 late payments + RS03 current arrears both driven by the same delinquency event).

11.2 Intervention Recommendations (interventions.py)

Maps signal patterns to specific intervention actions:

Scenari			
S_TEMPORARY_MISS	LOW	RS05 isolated delay	Soft contact, verify reason, note e
S_REPEATED_DELAYS	MEDIUM	RS01 >= 0.5, no RS05	Schedule RM meeting, review payment
S_DECLINING_INCOME	HIGH	RS07 >= 0.5 OR RS09 >= 0.5	Urgent income verification, financi
S_INCREASING_EXPOSURE	MEDIUM	RS12 >= 0.5 OR RS14 >= 0.5	Limit review, cross-sell assessment
S_SUSTAINED_HIGH_RISK	CRITICAL	Tier=Red, score >= 80	Credit committee referral, collecti
S_INSUFFICIENT_INFO	MEDIUM	DCS < 40	Request updated financials, initiat

11.3 Priority Scoring (priority.py)

Composite priority score (0-100) for watchlist ordering:

```
priorit
        + exposure x 0.25
        + probability x 0.25
        + trajectory x 0.15
        + correlation x 0.05
```

Where:

- severity = risk_score (0-100)
- exposure = normalised outstanding balance
- probability = delinquency forecast p_3m (preferred), or ML second opinion, or roll probability
- trajectory = extended trajectory weight (Sustained-Deteriorating=1.0 down to Sustained-Improving=0.10)
- correlation = combined_risk_factor x 100

Priority Quadrants: Classifies customers into a 2x2 matrix of High/Low Priority vs High/Low Severity for triage visualisation.

11.4 Trend Analysis (trends.py)

Historical Scoring:

- Re-scores the entire portfolio at 3-month intervals over the trailing 12 months
- Produces a time series of risk scores and module indices per customer

Extended Trajectory Classification:

- Uses OLS slope of historical scores + volatility (coefficient of variation)
- Categories: Sustained-Deteriorating, Accelerating, Volatile-Worsening, Volatile, Stable, Volatile-Improving, Improving, Sustained-Improving

Indicator Timeline:

- Extracts underlying metric time series (DPD, inflows, balance, utilisation) for a single customer
- Visualised as a faceted chart in the Investigation page

12. Regulatory Compliance (RBI SMA)

12.1 SMA Classification (rbi.py)

Implements the Reserve Bank of India's Special Mention Account framework:

Classifi		
Standard	0	No overdue
SMA-0	1-30	Principal/interest overdue 1-30 day
SMA-1	31-60	Principal/interest overdue 31-60 da
SMA-2	61-90	Principal/interest overdue 61-90 da
NPA	91+	Non-performing asset

12.2 Regulatory Reports

Report	Description
SMA Distribution	Count and exposure by SMA category
SMA by Portfolio	Breakdown across portfolio segments
SMA Migration Matrix	Period-over-period transition heatm
SMA Trend	12-month trend of SMA distribution
SMA-2/NPA Account List	Detailed list for regulatory filing
EWS Compliance Summary	At-risk counts, exposure, pre-delin

12.3 Export

Full regulatory pack exported as Excel workbook (.xlsx) with separate sheets per report, formatted for submission.

13. Storage & Audit Trail

13.1 DuckDB Schema (store.py)

```
-- Frozen alerts: immutable snapshot at publish time
CREATE TABLE alerts (
    alert_id        VARCHAR PRIMARY KEY,
    run_id          VARCHAR,
    customer_id     VARCHAR,
    published_at    TIMESTAMP,
    original_score  DOUBLE,
    original_tier   VARCHAR,
    confidence      VARCHAR,
    reasons_json    VARCHAR,
    evidence_json   VARCHAR
)

-- Append-only analyst decisions
CREATE TABLE analyst_decisions (
    decision_id     VARCHAR PRIMARY KEY,
    alert_id        VARCHAR,
    customer_id     VARCHAR,
    analyst         VARCHAR,
    decision        VARCHAR,  -- CONTINUE_MONITORING | REQUIRES_INVESTIGATION |
                              -- FALSE_POSITIVE | ESCALATE_FOR_REVIEW
    rationale       VARCHAR,
    decided_at      TIMESTAMP
)

-- Active interventions with status tracking
CREATE TABLE interventions (
    intervention_id VARCHAR PRIMARY KEY,
    customer_id     VARCHAR,
    scenario        VARCHAR,
    urgency         VARCHAR,
    actions_json    VARCHAR,
    status          VARCHAR,  -- OPEN | IN_PROGRESS | COMPLETED |
                              -- DEFERRED | ESCALATED
    assigned_to     VARCHAR,
    created_at      TIMESTAMP,
    updated_at      TIMESTAMP
)

-- Append-only intervention change log
CREATE TABLE intervention_history (
    history_id      VARCHAR PRIMARY KEY,
    intervention_id VARCHAR,
    old_status      VARCHAR,
    new_status      VARCHAR,
    changed_by      VARCHAR,
    notes           VARCHAR,
    changed_at      TIMESTAMP
)

-- Free-form investigation notes
CREATE TABLE investigation_notes (
    note_id         VARCHAR PRIMARY KEY,
    customer_id     VARCHAR,
    author          VARCHAR,
    content         VARCHAR,
    created_at      TIMESTAMP
)
```

13.2 Design Principles

- Alerts are frozen: Once published, an alert's original_score, original_tier, and evidence are never updated. Live scores may diverge, enabling before/after comparison.
- Decisions are append-only: Multiple decisions can exist per alert, creating a full decision history.
- Interventions track lifecycle: OPEN -> IN_PROGRESS -> COMPLETED/DEFERRED/ESCALATED, with full change history.

14. User Interface

14.1 Technology

- Framework: Streamlit
- Charts: Altair (declarative, interactive)
- Layout: Sidebar navigation with 10 pages

14.2 Page Inventory

Page 1: Home (Morning Briefing)

The analyst's starting page providing a portfolio-level overview:

- KPI Tiles: Red count, Amber count, new escalations, total at-risk exposure
- Forecast Metrics: Expected new 30+ DPD accounts (1m/3m), at-risk exposure from forecasts
- AI Briefing: LLM-generated or template morning briefing paragraph
- Red-Flag Tabs (5):

1. Likely 30+ DPD (silent stress from forecast)
2. Newly Escalated (tier increased this period)
3. Hard Triggers (DPD-based overrides)
4. Emerging Segments (statistically elevated clusters)
5. Recovering (previously Amber/Red, now improving)

- Delinquency Outlook Charts: Forecast distribution, probability vs current score scatter
- Model Card: Expandable panel with AUC, feature importance, validation methodology

Page 2: Upload & Validate

- Bundled Datasets: 4 pre-loaded options (Gulf Bank clean/dirty, Northern Emirates clean/dirty)
- File Upload: CSV/Excel/JSON upload with drag-and-drop
- Validation Dashboard: KPIs (total records, rejected, warnings), per-table summary cards, relationship integrity matrix, filterable issues table

Page 3: Portfolio Dashboard

- Tier Distribution: Stacked bar/donut showing Green/Watch/Amber/Red split
- Trajectory Breakdown: Counts by Deteriorating/Stable/Improving/Recovering
- Emerging Clusters: Table with segment, lift ratio, count, significance
- Roll-Rate Heatmap: Markov transition matrix visualised as colour-coded grid
- Pre-Delinquent vs Delinquent: Stacked breakdown showing early vs late warning effectiveness
- Biggest Movers: Customers with largest score changes
- Intervention Workload: Pending vs completed interventions by urgency
- Priority Matrix: Scatter plot of priority vs severity with quadrant labels
- Multi-Signal Correlation: Pattern frequency and overlap analysis
- Accuracy Panel: Ground-truth comparison (if labels available)

Page 4: Watchlist

- 9 Filters: Tier, product, segment, portfolio, DPD range, emirate, trajectory, confidence, SMA classification
- Sortable Table: Columns include priority score, risk score, tier badge, forecast probability, top signal, recommended intervention
- Actions: Click to open Customer Detail or Investigation page

Page 5: Customer Detail

- Score Breakdown: Tier badge, risk score gauge, confidence indicator
- Signal Points Chart: Horizontal bar showing each signal's point contribution
- Signal Table: All 21 signals with firing status, strength, weight, contribution
- Module Indices: RBDI, CFSI, LPI, BUREAU as percentage gauges
- AI Summary: LLM-generated or template risk narrative
- Forecast Panel: P(30+ DPD) for 1m/3m with top-3 SHAP drivers
- ML Second Opinion: P(deterioration 90d) with SHAP waterfall chart
- History Charts (4): DPD trend, income/inflow trend, utilisation trend, total obligations trend
- Detail Tabs (4): Repayment schedule, facility list, exposure snapshot, exception evidence

Page 6: Investigation

- Priority & SMA Metrics: Priority score, SMA classification, urgency level
- Correlated Deterioration: Multi-signal correlation analysis for this customer
- Module Index Chart: Radar or bar chart of RBDI/CFSI/LPI/BUREAU
- Recommended Interventions: Matched scenarios with expandable action lists
- Historical Scoring: 12-month risk score trend at 3-month intervals
- Indicator Timeline: Faceted chart showing DPD, inflows, balance, utilisation over time
- Case File: Active interventions (create/update), investigation notes (add/view)

Page 7: Analyst Review

- Publish Action: Freeze current alerts to DuckDB

- Alert List: Filterable by tier, published date, decision status
- Comparison View: Original (frozen) score vs current live score, highlighting divergence
- Decision Form: 4 options (Continue Monitoring, Requires Investigation, False Positive, Escalate for Review) with rationale text
- Decision History: Append-only timeline of all decisions per alert

Page 8: Rules Editor

- Signal Table: Editable grid for all 21 signals (toggle enabled, adjust threshold, severe, weight)
- Score Bands: Editable band boundaries
- Instant Re-score: Changes trigger immediate Stage 2 re-computation
- Before/After Counts: Shows impact of rule changes on tier distribution
- Reset: Restore factory defaults
- Download: Export current rules as JSON

Page 9: RBI Reporting

- SMA Distribution Chart: Bar chart of account/exposure by SMA category
- SMA by Portfolio: Grouped bar chart
- SMA Migration Matrix: Heatmap showing period transitions
- SMA Trend: 12-month line chart
- Account List: Detailed SMA-2/NPA accounts table
- Export: Download full regulatory pack as Excel

Page 10: AI Assistant

- Chat Interface: Multi-turn conversation with the LLM
- Context Grounding: All responses grounded in actual customer data and evidence
- Suggested Questions: Pre-built prompts for common analyst queries
- Session Memory: Maintains conversation context within the session

15. Data Generation & Simulation

15.1 Generator Architecture (datagen/)

Module	Purpose
`generate.py`	CLI entry point, orchestrates gener
`config.py`	Profile definitions, archetype dist
`rules.py`	Risk signal definitions, score band
`tables.py`	Table generation logic for all 8 da
`simulate.py`	Behavioural simulation per archetyp
`dirty.py`	Data quality defect injection
`checks.py`	Post-generation validation and stat

15.2 Generation Command

```
uv run python datagen/generate.py --profile northern_sme --customers 1200 --seed 2026 --out data/northern_sme
```

15.3 Archetype System

Individual Archetypes (11):

Archety		
STABLE	50%	Regular payments, steady income
ISOLATED_EXCEPTION	9%	One missed payment, then recovery
CHRONIC_LATE	4%	Frequent late payments, never cures
REPAYMENT_DETERIORATION	7%	Progressively worsening payments
INCOME_DECLINE	6%	Salary/income drops over time
OVER_LEVERAGED	6%	High utilisation, new credit seekin
ROLL_FORWARD	4%	Progressive DPD bucket migration
CURED	5%	Was delinquent, now recovered
THIN_FILE	5%	Minimal history, bureau no-hit
SALARY_ELSEWHERE	4%	Salary not visible at this bank
LEGACY_NPL	-	Already in 90+ DPD (Northern Emirat

SME Archetypes (6):

Archety		
SME_STABLE	48%	Healthy business, regular payments
SME_SEASONAL	15%	Cash-flow varies with business seas
SME_CASHFLOW_STRESS	14%	Declining business income
SME_OVER_LEVERAGED	8%	Excessive borrowing vs revenue
SME_REPAYMENT_DETERIORATION	7%	Worsening payment patterns
SME_THIN_FILE	8%	New business, limited history

15.4 Profile System

Parameter	Gulf Bank (default)	Northern Emirates
Customers	600	1,200
SME Ratio	30%	45%
Currency	AED	AED
Risk Clusters	2	3
NPL Rate	~0.2%	~5.6%
Stress Level	Normal	Heavy
LEGACY_NPL	No	Yes

15.5 Outputs

For each profile, the generator produces:

data/{p

```
  sample/
    csv/          # 8 clean CSV files
    excel/        # Single bundled Excel workbook
    rules/        # Signal definitions, score bands
  dirty/
    csv/          # 8 CSV files with injected defects
    excel/        # Bundled dirty Excel workbook
    _injected_issues.csv  # Ground truth of all injected defects
  _ground_truth/
```

```
    customer_labels.csv   # Expected risk bands per customer
  PROFILE.md              # Profile documentation
```

16. Technology Stack

Layer	Technology	Version	Purpose
Language	Python	3.12+	Core runtime
Package Manager	uv	Latest	Fast dependency management
UI Framework	Streamlit	Latest	Interactive web application
Charting	Altair	Latest	Declarative statistical visualisati
Data Processing	pandas, NumPy	Latest	Vectorised data manipulation
ML Framework	LightGBM	Latest	Gradient boosting for forecasts
Explainability	SHAP	Latest	Model interpretation
ML Utilities	scikit-learn	Latest	Preprocessing, metrics, CV
Imbalanced Learning	imbalanced-learn	Latest	SMOTE oversampling
Storage	DuckDB	Latest	Embedded analytical database
LLM Runtime	Ollama	Latest	Local LLM inference
LLM Model	qwen3:4b-instruct	4B params	Risk narrative generation

17. Performance & Validation Results

17.1 Clean Sample Results (Gulf Bank, 600 customers)

Metric	Value
Records ingested	67,897
Validation issues	0
Green	470 (78.3%)
Watch	47 (7.8%)
Amber	48 (8.0%)
Red	35 (5.8%)

17.2 Accuracy Metrics

Metric	Value	Description
High/Critical catch rate	75.8%	Customers expected Amber/Red correc
False alarm rate (low-risk)	0.0%	Low-risk customers incorrectly flag
Within one tier	97.5%	Predicted tier within 1 band of gro
Trap customers flagged	0/130	Zero false positives on planted tra

17.3 Forecast Model Performance

Horizon	AUC	Top-10% Capture
1 month	0.96	90%
3 months	0.89	67%

17.4 Dirty Data Handling

Metric	Value
Records rejected (ERROR)	516
Records warned (WARNING)	42
Defect types detected	15/15 (100%)
Injected issues	73

17.5 Emerging Cluster Detection

Cluster	Segment	Lift	Significance
Top cluster	Construction & Contracting, Sharjah	4.3x	High

18. Evolution & Commit History

Commit	Description
`3a533d6`	First commit - initial project setu
`9f95414`	Feature set definition
`532ead7`	Architecture diagram
`e9712fc` - `631750f`	Models and data architecture
`0cfae8f`	Synthetic EWS dataset generator
`81baad3`	Application architecture review
`f7137fe`	1-hour prototype plan
`755f532`	Sentinel EWS Streamlit prototype
`bc01280`	LLM bake-off (selected qwen3:4b-ins
`a5ae4c9`	Gitignore and LLM documentation
`91f796c`	**FINAL_ARCHITECTURE:** What-If Sim
`66da100`	New files for Northern Emirates dat
`1478621`	**ML Enhancement:** SMOTE, cross-va
`ed62ea7`	**Phase 2:** Risk correlation, inte

Appendix A: Risk Signal Quick Reference

```
REPAYMENT (RBDI)                    CASH-FLOW (CFSI)
  RS01  Late/missed       [30]        RS07  Inflow decline    [20]
  RS02  Worsening DPD     [15]        RS08  Consecutive drops [8]
  RS03  Current arrears   [15]        RS09  Salary stopped    [15]
  RS04  Partial payments  [10]        RS10  Balance erosion   [10]
  RS05  Isolated recovery [-10]       RS11  Returned payments [10]
  RS06  Recent cure       [8]

EXPOSURE (LPI)                      BUREAU
  RS12  High utilisation  [12]        RS17  External delinq   [12]
  RS13  Utilisation spike [10]        RS18  Bureau score drop [6]
  RS14  Debt burden       [15]
  RS15  New credit        [12]      DATA QUALITY (weight=0)
  RS16  Exp vs income     [10]        DQ01  Insufficient history
                                      DQ02  Missing financials
                                      DQ03  Bureau no-hit
```

Appendix B: Hard Trigger Rules

Override	Condition	Effect
OV01	DPD >= 90	Minimum tier = Red
OV02	DPD >= 30	Minimum tier = Amber
OV03	Low confidence + no hard evidence	Maximum tier = Watch

Appendix C: Module Index Formulas

```
RBDI   = (RS01 + RS02 + RS03 + RS04 + RS06 earned) / (RS01-RS04 + RS06 available) x 100
CFSI   = (RS07 + RS08 + RS09 + RS10 + RS11 earned) / (RS07-RS11 available) x 100
LPI    = (RS12 + RS13 + RS14 + RS15 + RS16 earned) / (RS12-RS16 available) x 100
BUREAU = (RS17 + RS18 earned) / (RS17-RS18 available) x 100
```

Document generated from codebase analysis on 2026-10-31. Covers all changes through commit ed62ea7 on branch feat/pranav.